

STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

A project by:-

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1. ABSTRACT

Educational data mining has emerged as an important research area for understanding and improving student learning outcomes. This project focuses on predicting student performance categories using machine learning techniques applied to a student performance dataset containing demographic, academic, behavioural, and lifestyle-related attributes. The dataset consisted of 3000 student records and 32 variables.

A complete machine learning pipeline was implemented, including data loading, identification of numerical and categorical variables, missing value treatment, label encoding, feature scaling, train-test splitting, model training, and feature importance analysis. A Decision Tree Classifier was used as the predictive model due to its interpretability and effectiveness for classification tasks.

Initial experimentation resulted in perfect classification accuracy because the feature **Final Exam Score** directly determined the target variable **Performance Category**. To obtain a more realistic assessment, an additional experiment was conducted by removing the Final Exam Score feature. The model then achieved an accuracy of approximately 53.33%, indicating that the remaining behavioral and academic factors possess moderate predictive power.

The project demonstrates the importance of data pre-processing, feature analysis, and critical evaluation of machine learning results. It also highlights the issue of target leakage, a common challenge in predictive modelling.

Keywords: Machine Learning, Educational Data Mining, Decision Tree Classifier, Student Performance Prediction, Feature Importance, Target Leakage.

2. INTRODUCTION

Machine learning has become a powerful tool for extracting insights from educational data. Educational institutions collect information regarding attendance, study habits, examination performance, lifestyle factors, and student engagement. Analysing these variables can help identify patterns associated with academic success and support evidence-based educational decision-making.

Student performance prediction is one of the most widely studied problems in educational data mining. Accurate prediction models can assist institutions in identifying students at risk of poor performance and enable timely interventions. Such predictive systems can contribute to improved academic outcomes and resource allocation.

This project investigates the application of supervised machine learning techniques to classify students into performance categories. The study uses a Decision Tree Classifier and follows a structured workflow consisting of data pre-processing, feature engineering, model training, evaluation, and interpretation.

Beyond achieving predictive accuracy, the project emphasizes model interpretability and critical assessment of results. Special attention is given to identifying variables that contribute most significantly to prediction performance.

3. PROJECT OBJECTIVE

The primary objective of this project is to develop a machine learning model capable of predicting student performance categories using academic, demographic, and behavioral attributes.

The specific objectives are:

- To load and explore the student performance dataset.
- To identify numerical and categorical features.
- To detect and handle missing values.
- To transform categorical variables into numerical representations.
- To normalize feature values using feature scaling techniques.
- To split the dataset into training and testing subsets.
- To train a Decision Tree Classifier for performance prediction.
- To evaluate model performance using classification accuracy.
- To analyze feature importance and understand the factors influencing prediction.
- To investigate the impact of target leakage on model performance.

4. METHODOLOGY

The project followed a standard machine learning workflow consisting of the following stages.

4.1 Data Collection and Loading

The Student Performance Dataset was obtained from Kaggle and loaded into Google Colab using the Pandas library.

Dataset characteristics:

- Total observations: 3000
- Total variables: 32
- Target variable: Performance Category

4.2 Data Exploration

The dataset was examined to identify:

- Feature types
- Dataset dimensions
- Missing values
- Target variable distribution

The features were categorized into:

Numerical Features (26)

Examples include:

- Study Hours Per Day
- Attendance Percentage
- Focus Score
- Sleep Hours
- Final Exam Score

Categorical Features (6)

Examples include:

- Gender
- City Type

- Mental State
- Learning Style
- Career Goal
- Performance Category

4.3 Missing Value Treatment

Missing values were detected in:

Variable	Missing Values
Social Media Hours	60
Focus Score	60
Sleep Hours	60

The following imputation techniques were applied:

- Numerical variables → Median Imputation
- Categorical variables → Mode Imputation

After pre-processing, no missing values remained in the dataset.

4.4 Label Encoding

Since machine learning algorithms require numerical input, all categorical variables were encoded using LabelEncoder.

The encoded variables included:

- Gender
- City Type
- Mental State
- Learning Style
- Career Goal
- Performance Category

4.5 Feature-Target Separation

The dataset was divided into:

- Features (X)

- Target Variable (y)

The resulting dimensions were:

- $X = (3000, 31)$
- $y = (3000,)$

4.6 Train-Test Split

The dataset was split into:

- Training Data: 80%
- Testing Data: 20%

Parameters:

- `test_size = 0.20`
- `random_state = 42`

Resulting dimensions:

- $X_{\text{train}} = (2400, 31)$
- $X_{\text{test}} = (600, 31)$

4.7 Feature Scaling

MinMaxScaler was applied to normalize feature values between 0 and 1.

This ensured that all variables contributed on a comparable scale during model training.

4.8 Model Development

A Decision Tree Classifier was selected because:

- It is interpretable.
- It handles nonlinear relationships effectively.
- It provides feature importance measures.
- It requires minimal assumptions regarding data distribution.

The model was trained using the scaled training dataset and evaluated on unseen testing data.

5. DATA ANALYSIS AND RESULTS

5.1 Missing Value Analysis

The dataset contained a total of 180 missing values distributed across three variables:

- Social Media Hours
- Focus Score
- Sleep Hours

After median imputation, all missing values were successfully removed.

5.2 Model Performance

Experiment 1: Original Dataset

The Decision Tree Classifier was trained using all available features.

Result:

Accuracy Score = 1.00

While this result appears impressive, further investigation was necessary because perfect accuracy is uncommon in practical machine learning applications.

5.3 Feature Importance Analysis

Feature importance scores revealed the following:

Feature	Importance
Final Exam Score	1.00
All Remaining Features	0.00

This finding indicated that the model relied entirely on Final Exam Score for classification.

5.4 Identification of Target Leakage

The perfect accuracy score suggested the presence of target leakage.

Target leakage occurs when a feature contains information that directly determines the target variable. In this dataset, Performance Category appears to be derived directly from Final Exam Score.

As a result, the model was not learning complex patterns from multiple variables; it was simply reproducing information already embedded within the Final Exam Score feature.

5.5 Experiment 2: Removing Final Exam Score

To obtain a more realistic assessment, Final Exam Score was removed from the feature set and the model was retrained.

Result:

Accuracy Score = 0.5333

This experiment demonstrated that student performance prediction becomes significantly more challenging when the dominant feature is removed.

5.6 Feature Importance After Leakage Removal

The most important variables became:

Feature	Importance
Study Hours Per Day	0.082
Procrastination Index	0.069
Consistency Score	0.056
Social Media Hours	0.050
Stress Level	0.050

These results are educationally meaningful because they highlight behavioral and academic habits associated with student performance.

5.7 Discussion

The project demonstrates the importance of critically evaluating machine learning results rather than relying solely on accuracy metrics.

The initial accuracy score of 100% appeared excellent but was largely caused by target leakage. After removing the problematic feature, the model achieved

more realistic performance and identified meaningful predictors of academic success.

This analysis illustrates a key principle in statistical learning: model interpretation and data understanding are often as important as predictive accuracy.

6. CONCLUSION

This project successfully implemented a complete machine learning pipeline for student performance prediction using a Decision Tree Classifier.

The study involved data pre-processing, missing value treatment, feature encoding, feature scaling, train-test splitting, model development, and feature importance analysis. The initial model achieved perfect accuracy due to the presence of a highly informative feature, Final Exam Score. Subsequent analysis revealed that this feature directly influenced the target variable, resulting in target leakage.

After removing the leakage source, the model achieved a more realistic accuracy of approximately 53.33%, demonstrating the predictive influence of variables such as study habits, consistency, procrastination, and stress levels.

The project highlights the importance of careful data analysis, feature interpretation, and validation in machine learning research. It also demonstrates practical applications of data science techniques in educational analytics and provides a foundation for future research involving more sophisticated predictive models.

7. APPENDICES

Appendix A: Software and Tools

- Python 3
- Google Colab
- Pandas
- NumPy
- Scikit-learn

Appendix B: Machine Learning Pipeline

1. Data Loading
2. Data Exploration
3. Missing Value Treatment
4. Label Encoding
5. Feature-Target Separation
6. Train-Test Split
7. Feature Scaling
8. Decision Tree Training
9. Prediction
10. Accuracy Evaluation
11. Feature Importance Analysis

Appendix C: Key Results

Metric	Result
Total Records	3000
Total Features	32
Missing Values Removed	180
Training Samples	2400
Testing Samples	600
Accuracy (Original Dataset)	1.00
Accuracy (Without Final Exam Score)	0.5333